



JAIPUR NATIONAL UNIVERSITY

Semester/Year End (Main/Back) Examinations, May-2025

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Computer Science) / B.Tech. CSE (Hons.) in Cyber Security / B.Tech. CSE (Hons.) in Artificial Intelligence and Machine Learning
Semester/Year:	Semester-IV
Paper Code:	BTECSVA401T22
Title of Paper:	Introduction to Machine Learning
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section-A and Section-B. Section-A consists of seven questions and Section-B consists of six questions. Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question Number	SECTION-A	Marks Allotted
Q.1	What is machine learning? Explain any two types of machine learning with suitable examples.	4
Q.2	Define CNF and DNF. How are Boolean functions represented using CNF?	4
Q.3	What is a threshold logic unit? Mention its role in neural networks.	4
Q.4	Compare neural networks with knowledge-based systems.	4
Q.5	.What is the purpose of decision trees in machine learning? Briefly explain their working.	4
Q.6	Explain the difference between inductive logic programming and decision tree induction.	4

Q.7 What is clustering in machine learning? Give two examples of clustering algorithms.

4

SECTION-B

Q.1 Explain the process of learning input-output functions in machine learning. Describe how decision lists are used for Boolean functions. 10

Q.2 Describe the architecture and working of feedforward neural networks. How does the backpropagation algorithm train these networks? 10

Q.3 What are belief networks? Describe the statistical learning method for learning belief networks. 10

Q.4 Explain the working of the nearest neighbor algorithm. What are its strengths and limitations in classification tasks? 10

Q.5 What is Inductive Logic Programming (ILP)? Explain how recursive programs are handled in ILP. 10

Q.6 What is computational learning theory? Explain the fundamental theorem of learning and give an overview of reinforcement and explanation-based learning. 10